

Motif finding in DNA sequences

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Introduction

In biology a motif is a short, recurring pattern in DNA, RNA or protein sequences, that often represents a specific biological function. The problem of finding motif is based on analyzing the characteristic patterns within the observed sequences.

In our project we consider a model of generating DNA sequences from alphabet letters $\Sigma = \{A, C, G, T\}$. Each sequence can come from one of the two models:

- motif model - described by the probability matrix Θ ,
- background model - described by the distribution Θ_{bg} .

Data used in this project comes from file `motif_data_known_alpha.json`. It consists of k DNA sequences of equal length w . Nucleotides are encoded as integers according to convention $A = 1, C = 2, G = 3, T = 4$. The sequences are stored in a matrix X , where each row corresponds to a single sequence. It also consist the value of α .

The motif model

Θ is a matrix of size $4 \times w$, where w is the length of a motif. Its rows correspond to letters A, C, G, T (encoded as 1,2,3,4) and its columns correspond to positions $1, \dots, w$. For example:

$$\Theta = \begin{pmatrix} 0.25 & 0.1 & 0.2 & 0.3 \\ 0.25 & 0.2 & 0.3 & 0.2 \\ 0.25 & 0.3 & 0.2 & 0.1 \\ 0.25 & 0.4 & 0.3 & 0.4 \end{pmatrix}$$

Each column of Θ describes the probability distribution of nucleotides A, C, G, T at a given motif position. From this, the sum of each column is equal to 1.

$\Theta_{a,j}$ is a probability of occurring a nucleotide on j -th position in the sequence generated by motif model. For example: $\Theta_{2,3} = 0.3$ means, that the probability of C occurring on the third motif position is equal to 0.3.

If a sequence $x = (x_1, \dots, x_w)$ is generated from the motif model, the probability of observing this sequence is given by:

$$\mathbb{P}(x|\Theta) = \prod_{j=1}^w \Theta_{x_j,j}.$$

For example, consider the sequence $x = (A, T, T, T)$. Using our example matrix, the probability that this sequence is generated by the motif model is $\mathbb{P}((A, T, T, T)|\Theta) = \Theta_{1,1} \cdot \Theta_{4,2} \cdot \Theta_{4,3} \cdot \Theta_{4,4} = 0.25 \cdot 0.4 \cdot 0.3 \cdot 0.4 = 0.012$.

The background model

$\Theta_{bg} = (\theta_A, \theta_C, \theta_G, \theta_T)$, is a vector consisting of global probability of occurring of each nucleotide. Its entries also sum to 1. In contrast to the motif model, the background model does not depend on the position in the sequence. From this, the same nucleotide probabilities are used at every position. For example: $\Theta_{bg} = (0.4, 0.1, 0.1, 0.4)$ means that A and T occur generally with probability 0.4, while C and G occur with probability 0.1, on every possible position. So if a sequence is generated from the background model, the probability of observing A at any position is always equal to 0.4, regardless of whether it is a first, second or any other position in a sequence. From this the probability of a sequence is obtained by multiplying the probabilities of its nucleotides:

$$\mathbb{P}(x|\Theta_{bg}) = \prod_{j=1}^w \Theta_{bg}(x_j)$$

For example, consider the sequence $x = (A, T, T, T)$. Using our exemplary background distribution, the probability of observing this sequence is $\mathbb{P}((A, T, T, T)|\Theta_{bg}) = 0.4 \cdot 0.4 \cdot 0.4 \cdot 0.4 = 0.0256$.

Generating sequences

The probability of generating a sequence from the motif model is equal to α and from the background - $1 - \alpha$. Assume that our sequences $x_1, \dots, x_k$ were generated as follows:

- For each $i = 1, \dots, k$ we independently simulate $Z_i \in \{0, 1\} : \mathbb{P}(Z_i = 1) = \alpha, \quad \mathbb{P}(Z_i = 0) = 1 - \alpha$.
- If $Z_i = 1$, we generate x_i from the motif model.
- If $Z_i = 0$, we generate x_i from the background distribution.

Main task of our project

As before we observe sequences $x_1, \dots, x_k$, but the variable Z_i is unknown, so we do not know which model generated a particular sequence. Therefore, the probability of observing a sequence x_i is given by a mixture of the two models:

$$\alpha P(x_i | \Theta) + (1 - \alpha) P(x_i | \Theta_{bg}).$$

The first part corresponds to the probability that the sequence was generated from the motif model. The second part corresponds to the probability that it was generated from the background model.

Assuming that all sequences are generated independently, the likelihood function for the entire dataset $X = x_1, \dots, x_k$ can be written as

$$L(\Theta) = \prod_{i=1}^k (\alpha P(x_i | \Theta) + (1 - \alpha) P(x_i | \Theta_{bg})).$$

More specifically, we aim to estimate $\hat{\Theta} = \operatorname{argmax}_{\Theta} L(\Theta) = \operatorname{argmax}_{\Theta} \mathcal{L}(\Theta)$, where $\mathcal{L}(\Theta) = \log L(\Theta)$. We use logarithm, because the logarithm transforms products into sums and improves numerical stability.

Because Z_i are unknown, direct maximization of the likelihood function is difficult. Therefore, the Expectation-Maximization (EM) algorithm is used to iteratively estimate the unknown parameters.

Expectation-Maximization algorithm

The EM (Expectation-Maximization) algorithm is an iterative method for estimating parameters of probabilistic models containing unknown variables. It is used, when direct maximization of the likelihood function is difficult due to uncomplete information about the data generation.

In our project the unknown variables are Z_i , which determine, whether a given sequence was generated from the motif model or the background model. Because we do not know them, we are unable to assign individual sequences to one of our models.

The EM algorithm solves this problem by recursively doing two steps: expectation step (E) and maximization step (M). In expectation step we calculate the probability of each sequence coming from the motif model. Next in the maximization step we update our model's parameters based on the calculated probabilities earlier. We continue the process until reaching convergence, which means that further update of parameters does not cause a significant change in value of log-likelihood function.

To estimate parameters we use data in file `motif_data_known_alpha.json`. Based on sequences in this file we want to estimate parameters of motif model Θ and the background model Θ_{bg} .

Implementation

At the beginning we load our matrix X and parameter α . We then read the number of sequences k and their length w . Because we do not know the parameters of Θ and Θ_{bg} we start our algorithm from random matrices. For each column of Θ we generate a random distribution of probabilities. Analogically, we randomly generate a vector Θ_{bg} .

E step

For each sequence we calculate the probability, that it was generated from the motif model $P(x_i|\Theta)$ and from the background model $P(x_i|\Theta_{bg})$. In the next step we use Bayesian formula:

$$\gamma_i = P(Z_i = 1|x_i) = \frac{\alpha P(x_i|\Theta)}{\alpha P(x_i|\Theta) + (1 - \alpha)P(x_i|\Theta_{bg})}.$$

γ_i is a probability, that the sequence comes from the motif model.

M step

After calculating all γ_i we update parameters of our model. For each j position and for each nucleotide a we calculate the weighted number of its occurrences:

$$\Theta_{a,j} = \frac{\sum_{i=1}^k \gamma_i \mathbf{1}(x_{ij} = a)}{\sum_{i=1}^k \gamma_i}.$$

Analogically we update the values of Θ_{bg} , but with use of weights $1 - \gamma_i$. Based on this, sequences that have high probability of coming from the motif model have more influence on estimation of Θ parameters. The sequences with low probability of being generated from the motif model have influence on Θ_{bg} parameters.

In addition during the parameter update, a small positive constant $\varepsilon = 10^{-8}$ is added to every count before normalization. Without this, if for example nucleotide G never appears on j position, then $\Theta_{G,j} = 0$. This can later lead to numerical problems (for example in logarithms), but adding a small constant solves this problem.

Log-likelihood

After updating parameters we calculate log-likelihood value:

$$\mathcal{L}(\Theta) = \sum_{i=1}^k \log(\alpha P(x_i|\Theta) + (1 - \alpha)P(x_i|\Theta_{bg})).$$

If the change of this value between next iterations is smaller than our threshold $1e^{-6}$ we stop the algorithm. At this point we assume that the algorithm has converged and further updates would not significantly improve the parameter estimates.

Multiple repetitions

The EM algorithm's flaw is that it can converge to the local maximum of the likelihood function. So to improve chances of finding a better result, we repeat the whole process 20 times, each time with random starting matrices. At the end from all our results we choose the one with the highest log-likelihood value.

Why the highest log-likelihood value and not the smallest?

We calculate probabilities, so

$$0 < P(x_i|\Theta), P(x_i|\Theta_{bg}) < 1.$$

From this

$$\log(P(x_i|\Theta)), \log(P(x_i|\Theta_{bg})) < 0.$$

So for example our calculated log-likelihood values may look like this:

$$ll_1 = -1270 \quad ll_2 = -1189 \quad ll_3 = -1212 \quad ll_4 = -1150 \quad \dots$$

Although all values are negative, a larger log-likelihood value corresponds to a better fit of the model to the observed data. Therefore, among the values above, the best result is $ll_4 = -1150$, because it is the largest value.

We start with $\text{best_ll} = -\infty$ and our estimated Θ, Θ_{bg} as empty matrices. After each run of the EM algorithm, we compare the obtained log-likelihood value with the current best value. If the new value is greater than best_ll , we update best_ll and save obtained parameters to our Θ, Θ_{bg} matrices. At the end the algorithm returns the final estimators $\hat{\Theta}, \hat{\Theta}_{bg}$.

Results with known alpha

In our case the obtained best_ll was equal to -10448.072. The parameter α was in the data and equal to 0.45. The resulting estimates of the motif matrix Θ and the background distribution Θ_{bg} were saved to the output file `estimated_params.json`. Below there are obtained results:

$$\hat{\Theta} = \begin{pmatrix} 0.59 & 0.10 & 0.10 & 0.17 & 0.51 & 0.10 & 0.17 & 0.20 \\ 0.18 & 0.66 & 0.11 & 0.10 & 0.15 & 0.57 & 0.12 & 0.15 \\ 0.13 & 0.14 & 0.61 & 0.13 & 0.20 & 0.18 & 0.56 & 0.12 \\ 0.10 & 0.10 & 0.18 & 0.60 & 0.14 & 0.15 & 0.15 & 0.53 \end{pmatrix}$$

$$\hat{\Theta}_{bg} = (0.3143, 0.2019, 0.2137, 0.2702)$$

The obtained motif matrix shows, that nucleotides have preferences at chosen individual places. From this, we conclude, that the EM algorithm identified successfully the motif structure in the data.

Results with unknown alpha

In the bonus part of the project, the parameter α is not in the provided file. We must estimate it additionally to the Θ and Θ_{bg} . We start with $\alpha = 0.5$ and after computing probabilities γ_i in E-step we obtain the estimator:

$$\hat{\alpha} = \frac{1}{k} \sum_{i=1}^k \gamma_i.$$

From this, α is estimated as the average probability that a given sequence comes from the motif model. After that, the new $\hat{\alpha}$ is then used in next EM-step with updated motif and background models.

In our case the best $_ll=-15855.24$ and $\hat{\alpha} \approx 0.37$. The obtained Θ and Θ_{bg} are as follow:

$$\hat{\Theta} = \begin{pmatrix} 0.64 & 0.08 & 0.15 & 0.09 & 0.47 & 0.11 & 0.14 & 0.19 \\ 0.15 & 0.66 & 0.08 & 0.18 & 0.19 & 0.53 & 0.10 & 0.13 \\ 0.14 & 0.16 & 0.67 & 0.14 & 0.24 & 0.16 & 0.60 & 0.22 \\ 0.07 & 0.10 & 0.10 & 0.60 & 0.10 & 0.19 & 0.16 & 0.46 \end{pmatrix}$$

$$\hat{\Theta}_{bg} = (0.31, 0.20, 0.20, 0.29)$$

The estimated $\hat{\alpha}$ is equal to 0.37, which means that approximately 37% of the sequences are expected to come from the motif model, while remaining 63% come from the background model.